

0/24 Questions Answered

hw8-diff-means-regression

Student Name

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Q1**1 Point**

Beginning with the introduction of the typewriter in the 1800's, an individual's ability to type quickly and accurately became a marketable skill. People used to take typesetting classes and research and training were conducted to improve how people learn the skill of typing. Today, proficient typing ability is commonly assumed to be a necessity, but differences in education and availability of personal devices has resulted in differences in typing ability for students across the world.

Pinet et al. ([2022](#)) conducted a study to investigate characteristics related to the modern students' typing proficiency at a large public university in France. This data set includes typing speed (words per minute) and typing accuracy (computed as the difference between the total number of characters in the target sentence and the number of insertions, deletions, substitutions, divided by the total number of characters in the target sentence).

We will use this data to investigate differences in a person's typing speed based on their typing accuracy.

	Typing speed $\bar{x}$	Typing speed s	n
Students with low typing accuracy	66.5963	16.9340	612
Students with high typing accuracy	63.5706	18.0786	689

We will consider a high typing accuracy to be an accuracy rate above 85% and low accuracy to be an accuracy of 85% or less. We want to investigate if there is a difference in mean typing speed by low versus high typing accuracy students. Let

- μ_1 = mean typing speed for someone with low typing accuracy
- μ_2 = mean typing speed for someone with high typing accuracy

Q1.1

1 Point

Is this an example of paired or unpaired data?

Unpaired

Paired

Save Answer

Q2**7 Points**

We will conduct a hypothesis test to investigate if there is evidence of a difference in mean typing speed between students with low and high typing accuracy, at a significance level of 0.05.

Q2.1**1 Point**

Select the appropriate hypotheses.

$$H_0 : \mu_1 - \mu_2 \neq 0$$

$$H_1 : \mu_1 - \mu_2 = 0$$

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_1 : \mu_1 - \mu_2 \neq 0$$

$$H_0 : \mu_2 - \mu_1 \neq 0$$

$$H_1 : \mu_2 - \mu_1 = 0$$

$$H_0 : \mu_2 - \mu_1 = 0$$

$$H_1 : \mu_2 - \mu_1 \neq 0$$

Save Answer

Q2.2**1 Point**

What is the point estimate for the difference in population mean typing speed between low and high typing accuracy students? Provide your final answer, rounding to four decimal places (make sure to keep the order of groups in mind).

Save Answer

Q2.3**1 Point**

What is the standard error of the point estimate? Round to four decimal places. Upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q2.4****1 Point**

Our next step is to calculate the p-value for our test. We know the difference of means is approximately t-distributed, but what are the degrees of freedom of this t-distribution?

$$df = 612 - 1 = 611$$

$$df = 689 - 1 = 688$$

$$df = 612 + 689 = 1301$$

Save Answer**Q2.5****1 Point**

Calculate the p-value and round to 4 decimal places. Upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer**

Q2.6**2 Points**

You should have found a very small p-value, much smaller than the significance level. Specify whether we reject the null hypothesis in support of the alternative hypothesis or if we fail to reject the null hypothesis, and then provide a conclusion in context based on the p-value.

Save Answer

Q3**3 Points**

Now we are not just interested in investigating if there is a difference, we actually want to estimate the difference in means. Let's calculate a 95% confidence interval to express our uncertainty in our point estimate.

Q3.1**1 Point**

What is the critical value for a 95% confidence interval for a mean? Round to four decimal places.

[Save Answer](#)**Q3.2****1 Point**

What is the lower bound of the 95% confidence interval for the difference of means? Round to four decimal places. Upload a picture of your work.

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

Q3.3**1 Point**

What is the upper bound of the 95% confidence interval for the difference of means? Round to four decimal places. Upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer**

Q4**5 Points**

In this homework we conducted a hypothesis test to test if there was evidence of a difference between the two population means. Then we calculated a confidence interval to estimate the difference between the population means.

What if instead we just want to study the relationship between typing speed and typing accuracy for students, where accuracy is a continuous variable (instead of discretized into low and high)?

Let's review simple linear regression:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Identify each component of the equation by selecting its definition.

Q4.1**1 Point**

x_i is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

Q4.2**1 Point** y_i is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

Q4.3**1 Point** ϵ_i is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

Q4.4**1 Point** β_1 is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

Q4.5**1 Point** β_0 is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

Q5**4 Points**

In our case, the variable we want to learn about is typing speed and typing accuracy is the explanatory variable. We wanted to understand if there is a linear relationship between typing speed and accuracy, at a significance level of 0.05.

Q5.1**1 Point**

What is the correct set of hypotheses to test if the mean typing speed linearly varies by accuracy?

$$H_0 : \beta_0 = 0$$

$$H_a : \beta_0 \neq 0$$

$$H_0 : \beta_0 \neq 0$$

$$H_a : \beta_0 = 0$$

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

$$H_0 : \beta_1 \neq 0$$

$$H_a : \beta_1 = 0$$

Save Answer

Q5.2

1 Point

Below we fit the simple linear regression model and show the output.

```
> typing_model <- lm(speed ~ accuracy, data = typing)
> broom::tidy(typing_model)
# A tibble: 2 × 5
  term          estimate std.error statistic  p.value
<chr>          <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)    82.9      6.69     12.4 2.14e-33
2 accuracy     -21.2      7.90     -2.68 7.43e- 3
```

What is the p-value for our test? Write your answer with three significant digits (recall you can enter 2.35×10^{-6} as `2.35e-6`).

Q5.3

1 Point

What would be an appropriate conclusion?

We reject the null hypothesis. We found evidence of a linear relationship between typing speed and accuracy for students.

We fail to reject the null hypothesis. We found evidence of a linear relationship between typing speed and accuracy for students.

We reject the null hypothesis. We did not find evidence of a linear relationship between typing speed and accuracy for students.

We failed to reject the null hypothesis. We did not find evidence of a linear relationship between typing speed and accuracy for students.

Q5.4

1 Point

We found evidence of a linear relationship, but let's provide an estimate with uncertainty of how much we expect typing speed to differ for a one unit increase in accuracy.

```
> confint(typing_model)
              2.5 %      97.5 %
(Intercept) 69.75763 96.007990
accuracy    -36.68122 -5.681756
```

What is the 95% confidence interval for the slope of the regression? Round to four decimal places.

Lower bound:

Upper bound:

Save Answer

Q6**3 Points**

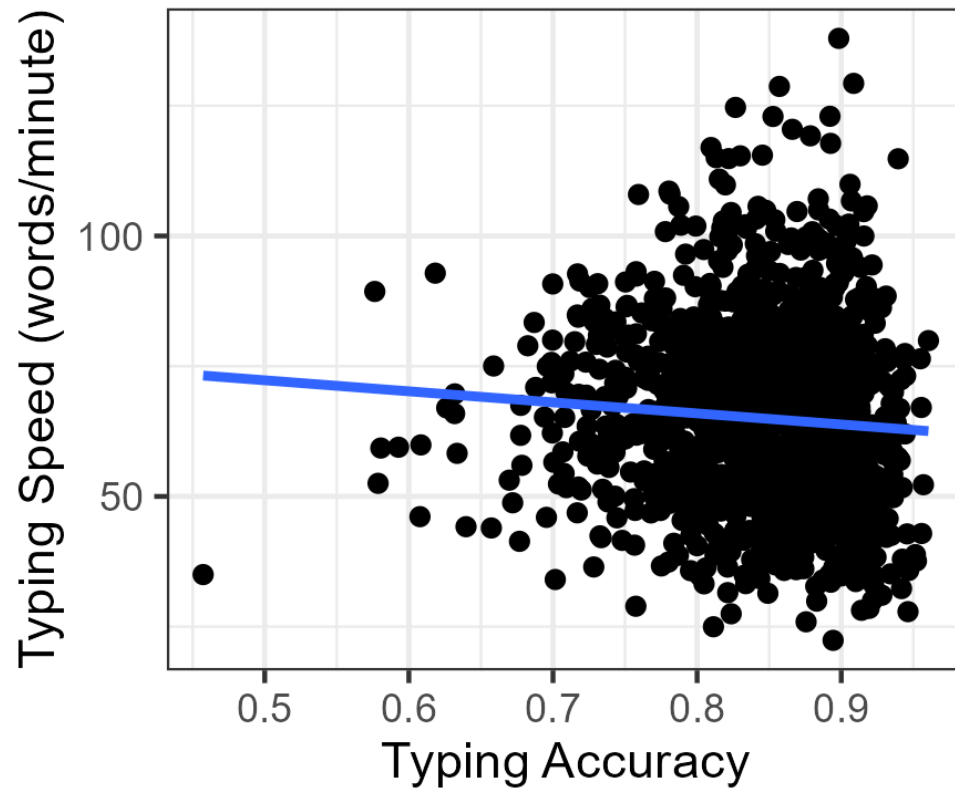
Two key things that linear regression is helpful for are:

- Investigating the expected difference in the response variable when the explanatory variable changes (through the estimate and confidence interval of β_1). *This is what we did in question 5.*
- Predicting the value for the response variable, given a value for the explanatory variable. *This is what we will do in question 6.*

Q6.1

1 Point

The linear regression line we fit is plotted below over the data.



Specify our estimated equation for the regression line (hint: look at the output shown in question 5.2):

[Save Answer](#)

Q6.2**1 Point**

Say a student has a typing accuracy of $0.6 = 60\%$. What would we estimate their typing speed to be? Enter your numeric answer and upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q6.3****1 Point**

Increasing a student's typing accuracy will decrease their typing speed.

True

False

We can't say from this analysis

Save Answer

Q7**1 Point**

The following question is related to the data topic we got to learn about, typing speed. There are no correct answers, this question is only being graded for completeness (note that answering "It was interesting" is an insufficient response).

Please explain in detail what you liked and/or what you did not like about this data topic.

Save Answer

Q8**0 Points**

- Have you done the homework by yourself, with a peer(s), or used tools like ChatGPT? If you do not acknowledge any help from a person or ChatGPT, we will consider this cheating.

- If you provided help to a peer(s) and/or got help from your peers(s) please explain briefly what this entailed.

- If you used ChatGPT, include the prompts you used and explain how ChatGPT supported your learning.

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